Kwanho Kim

SOFTWARE ENGINEER · C++ DEVICE SOFTWARE & INTEGRATION

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C++ engineer on semiconductor inspection equipment at KLA since 2026; previously robotics integration at Pickit 3D and V2X prototyping at Bosch. Works on serial device control, communication failures and logging for field diagnosis.

EXPERIENCE

KLA C++ Engineer — Aug 2026 – Present · Leuven, Belgium

- Designing fault-recovery logic for the equipment's lift mechanism and implementing it in C++, with team review (in progress)
- Made the camera's reported **white-balance settings** visible in the UI.
- Refactored **serial control and logging** into reusable components.

Pickit 3D Research & Software Engineer — Aug 2025 – Mar 2026 · Leuven, Belgium

- Resolved a **field communication failure** caused by a port conflict.
- Built a timezone-aware **log download and conversion workflow** with a multilingual UI for incident investigation.
- Built a reusable **Python client** for integrating internal tools, with reconnection and message checks; contributed to interface design.

Bosch V2X Communication Intern — Jun 2024 – Nov 2024 · Hildesheim, Germany

- Built a **V2X demonstrator** for safety scenarios; tested end-to-end messaging.
- Built the **operator interface** for an eBike safety prototype in React.

PROJECTS

WINNER **1st** — **Color Saves Life** — Feb 2024

digital.auto & mobility challenge · BCX 2024

Built the Python client for a driver visual-assistance prototype, integrated eye tracking with simulation and presented the team's demo.

TECHNICAL SKILLS

Languages	C++ (primary), C, Python; TypeScript / JavaScript, Bash
Device software	Serial device control, logging and diagnostics; fault-recovery design; CAN (basic)
Integration	REST, WebSocket, D-Bus, ROS 1/2; Docker; CARLA simulation
Operator UIs	Qt/QML, React
Tooling & AI	Git, Make; AI coding agents (Claude Code, Devin) for refactoring and tests

EDUCATION

Mechatronics · Bachelor's in progress · KNOU Prime College — 2026 – Present

Artificial Intelligence · Bachelor's degree awarded · Korea National Open University — 2025

Business Administration · Bachelor's degree awarded · Academic Credit Bank System — 2022

Computer Engineering · Bachelor's degree awarded · Academic Credit Bank System — 2020

TRAINING

42 · Software engineering · Seoul / Wolfsburg — Completed

SEA:ME · Automotive & Mobility · Wolfsburg — Jul 2023 – Jun 2024

Korean: Native · English: Fluent (IELTS 7.5, 2025) · Japanese: Learner